Heat engine

X21 (2021) — English translation

The Carnot heat engine operates with two heat reservoirs of finite heat capacity $C = 500$ Дж/К. which at the initial moment of time are at temperatures $T_h = 625$ К and $T_c = 298$ К. The power developed by a heat engine decreases exponentially with time $P = P_0\,\mathrm{e}^{-kt}$, where $P_0 = 960$ Вт is the power at the initial moment of time, and t is the time elapsed since the initial moment.

A1	At what temperature of the thermal reservoirs will the heat engine stop working?	**1.00**
A2	How much work will the heat engine do during its operation?	**1.00**
A3	After what time will the power developed by the heat engine be halved?	**1.00**
A4	After what time will the efficiency of a heat engine decrease by half?	**1.00**
A5	How long will it take for the temperature difference between the heat reservoirs to be halved?	**1.00**

Source: https://pho.rs/p/101